Average Hamiltonian Theory

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1 Conventions and Notation

1. Use $\hbar = 1$ unit system and d-dimensional Hilbert space $\mathcal{H}$ with $d < \infty$. The vector space of linear operators on $\mathcal{H}$ (Liouville space) is denoted by $\mathcal{L}(\mathcal{H})$.

2. Hilbert–Schmidt inner product on $\mathcal{L}(\mathcal{H})$ and corresponding norm:

$$\langle A, B\rangle \equiv \mathrm{Tr}\big(A^\dagger B\big), \qquad \|A\| \equiv \sqrt{\langle A, A\rangle} = \sqrt{\mathrm{Tr}(A^\dagger A)}.$$

3. Spin-I angular momentum operators: I_α with $\alpha \in \{x, y, z\}$.

4. Matrix product notation: $\prod_{n=1}^{M} A_n = A_M \cdots A_1$ and $\overrightarrow{\prod}_{n=1}^{M} A_n = A_1 \cdots A_M$.

5. Time-ordered exponential $\mathrm{Texp}\left[\int_{t_0}^{t} dt'\, G(t')\right]$ is defined as the solution $V(t)$ to the initial value problem $\dot{V}(t) = G(t)V(t)$ with $V(t_0) = \mathbb{1}$.

6. Natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ and whole numbers $\mathbb{W} = \{0, 1, 2, 3, \dots\}$.

7. Permutation group over n elements: S_n.

2 Interaction Frame and Dyson Series

- Consider the perturbation theory problem $H(t) = H_0(t) + \delta H(t) \in \mathcal{L}(\mathcal{H})$, where we know the solution for $H_0(t)$, and want to approximate the effect of adding $\delta H(t)$ to the propagator at a particular, fixed time T.
- Go to the interaction frame of $H_0(t)$:

$$\delta\tilde{H}(t) = U_0^\dagger(t)\,\delta H(t)\,U_0(t), \qquad U_0(t) = \mathrm{Texp}\left[-i\int_0^t dt'\,H_0(t')\right].$$

- *Claim:* The propagators in the two frames are

$$\begin{cases} \tilde{U}(t) = \mathrm{Texp}\left[-i\int_0^t dt'\,\delta\tilde{H}(t')\right] & \text{(interaction frame)} \\ U(t) = U_0(t)\tilde{U}(t) & \text{(original frame)} \end{cases}$$

  Proof: The state in the interaction frame satisfies

$$|\tilde{\psi}(t)\rangle = U_0^\dagger(t)|\psi(t)\rangle = U_0^\dagger(t)U(t)|\psi(0)\rangle = \underbrace{U_0^\dagger(t)U(t)}_{\tilde{U}(t)}|\tilde{\psi}(0)\rangle,$$

  where $U_0(0) = \mathbb{1}$. Hence, $U(t) = U_0(t)\tilde{U}(t)$.

2.1 Dyson Series

- Find $\tilde{U}(T)$ using Dyson series:

$$\begin{cases} \dfrac{d\tilde{U}(t)}{dt} = -i\delta\tilde{H}(t)\tilde{U}(t) \\ \tilde{U}(0) = \mathbb{1} \end{cases}$$

  Integrating gives

$$\tilde{U}(T) = \mathbb{1} - i\int_0^T dt'\,\delta\tilde{H}(t')\tilde{U}(t')$$

- Solve by iteration (Neumann series): Define the sequence

$$\begin{cases} \tilde{U}_n(T) = \mathbb{1} - i\displaystyle\int_0^T dt'\,\delta\tilde{H}(t')\tilde{U}_{n-1}(t'), & n \in \{2,3,\dots\} \\ \tilde{U}_1(T) = \mathbb{1} \end{cases}$$

  Such that $\tilde{U}(T) = \lim_{n\to\infty}\tilde{U}_n(T)$. So

$$\tilde{U}(T) = \mathbb{1} - i\int_0^T dt'\,\delta\tilde{H}(t') + (-i)^2\int_0^T dt_1\int_0^{t_1} dt_2\,\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) + \cdots$$

- Define the shorthand

$$\mathcal{D}_n(\delta\tilde{H}, T) \equiv \int_0^T dt_1 \cdots \int_0^{t_{n-1}} dt_n \prod_{j=1}^{n}\delta\tilde{H}(t_j), \qquad n \in \mathbb{N},$$

  and $\mathcal{D}_0(\delta\tilde{H}, T) \equiv \mathbb{1}$. The Dyson series is then

$$\boxed{\tilde{U}(T) = \sum_{n=0}^{\infty}(-i)^n\mathcal{D}_n(\delta\tilde{H}, T)}$$

1. *Lemma:* The nth term in the Dyson series is $\mathcal{O}\big(T^n\|\delta\tilde{H}\|_{\max}^n\big)$, where $\|\delta H\|_{\max} \equiv \max_{t\in[0,T]}\|\delta H(t)\| = \|\delta\tilde{H}\|_{\max}$ and the implicit constant in the $\mathcal{O}$ notation is only a function of n.

   Proof: For $n \geq 1$, we have

$$\begin{aligned}\|\mathcal{D}_n(\delta\tilde{H},T)\| &\leq \int_0^T dt_1 \cdots \int_0^{t_{n-1}} dt_n \|\delta\tilde{H}(t_1)\cdots\delta\tilde{H}(t_n)\| \qquad (\triangle \text{ inequality})\\ &\leq \int_0^T dt_1 \cdots \int_0^{t_{n-1}} dt_n \prod_{i=1}^n \|\delta\tilde{H}(t_i)\| \qquad (\text{submultiplicative norm})\\ &\leq \frac{T^n}{n!}\|\delta\tilde{H}\|_{\max}^n \qquad (\text{Appendix B})\end{aligned}$$

$$\Rightarrow \boxed{\|\mathcal{D}_n(\delta\tilde{H},T)\| \leq T^n\|\delta\tilde{H}\|_{\max}^n/n!, \qquad n\in\mathbb{N}}$$

- This gives us a physical criterion for when truncating the series is a good approximation. The time T has to be very short compared to the time-scale set by the perturbation Hamiltonian.

3 Magnus Series

- Define the interaction-frame *average Hamiltonian* $\bar{H}(T)$ for time T as the Hermitian operator

$$\bar{H}(T) \equiv \frac{i}{T}\ln\tilde{U}(T) = -\sum_{k=1}^d \frac{\theta_k(T)}{T}\,|u_k(T)\rangle\langle u_k(T)|\,,$$

  where

$$\tilde{U}(T) = \sum_{k=1}^d e^{i\theta_k(T)}\,|u_k(T)\rangle\langle u_k(T)|\,,$$

  is the spectral decomposition of the interaction-frame propagator. This definition ensures that $\tilde{U}(T) = e^{-iT\bar{H}(T)}$, i.e. *at time* T, the time evolution coincides with that of a fictitious static Hamiltonian $\bar{H}(T)$.

- The notion of an average generator is more general than just Hamiltonian operators, and it can be applied to, e.g., Liouville-space dynamical maps for open quantum systems (see Appendix A).

- Finding $\bar{H}(T)$ exactly is usually challenging. Instead, we develop a perturbative series for it in $\delta\tilde{H}(t)$, called the Magnus series.

- Advantages over a Dyson series:

  1. The resulting propagator is unitary at any order after truncation.
  2. Simplifies the analysis conceptually through the notion of an average Hamiltonian.

3.1 Constructing the Expansion

- Change $\delta\tilde{H}(t) \to \lambda\delta\tilde{H}(t)$ for bookkeeping, and expand the average Hamiltonian as a power series in λ:

$$\bar{H}(T,\lambda) = \sum_{m=0}^\infty \lambda^{m+1}h_m(T).$$

  The labels start at zero following magnetic resonance convention. The propagator is

$$\begin{aligned}\tilde{U}(T,\lambda) &= e^{-iT\sum_{m=0}^\infty \lambda^{m+1}h_m(T)}\\ &= \mathbb{1} + \sum_{n=1}^\infty (-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H},T).\end{aligned}$$

- Taking the logarithm of the propagator and using its Taylor series gives

$$\begin{aligned}-iT\sum_{m=0}^{\infty}\lambda^{m+1}h_m(T) &= \ln\left(\mathbb{1}+\sum_{n=1}^{\infty}(-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H},T)\right)\\ &= \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}\left(\sum_{n=1}^{\infty}(-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H},T)\right)^k\\ &= \sum_{k=1}^{\infty}\sum_{n_1=1}^{\infty}\cdots\sum_{n_k=1}^{\infty}\frac{(-1)^{k+1}}{k}(-i)^{\sum_{l=1}^{k}n_l}\lambda^{\sum_{l=1}^{k}n_l}\prod_{l=1}^{k}\mathcal{D}_{n_l}(\delta\tilde{H},T)\end{aligned}$$

  Now equate terms of order λ^{m+1}:

$$-iTh_m(T) = (-i)^{m+1}\sum_{k=1}^{m+1}\frac{(-1)^{k+1}}{k}\sum_{\substack{n_1+\cdots+n_k=m+1\\ n_1,\ldots,n_k\in\mathbb{N}}}\prod_{l=1}^{k}\mathcal{D}_{n_l}(\delta\tilde{H},T).$$

  The inner sum is over all $\{n_j\}_{j=1}^k$ configurations with $n_j\in\mathbb{N}$ and $\sum_{j=1}^k n_j = m+1$. Thus, for $m\in\mathbb{W}$, the mth Magnus term is

$$\boxed{h_m(T) = \frac{(-i)^m}{T}\sum_{k=1}^{m+1}\frac{(-1)^{k+1}}{k}\sum_{\substack{n_1+\cdots+n_k=m+1\\ n_1,\ldots,n_k\in\mathbb{N}}}\prod_{l=1}^{k}\mathcal{D}_{n_l}(\delta\tilde{H},T)}. \tag{1}$$

3.2 First Few Magnus Terms

- <u>*Zeroth-order term:*</u> For $m=0$, only one configuration exists, which has $k=1$ and $n_1=1$, giving

$$\boxed{h_0(T) = \frac{1}{T}\mathcal{D}_1(\delta\tilde{H},T) = \frac{1}{T}\int_0^T dt\,\delta\tilde{H}(t)}.$$

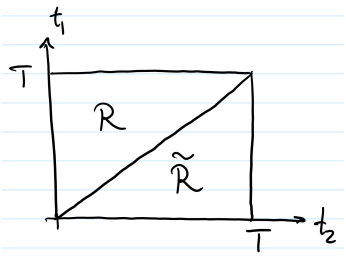
- <u>*First-order term:*</u> For $m=1$, the configurations are $n_1=2$ for $k=1$ and $n_1=n_2=1$ for $k=2$. Note that we can use the shorthand (2) and $(1,1)$ to enumerate the configurations. So

$$h_1(T) = -\frac{i}{T}\left[\mathcal{D}_2(\delta\tilde{H},T) - \frac{1}{2}\mathcal{D}_1^2(\delta\tilde{H},T)\right].$$

  Writing out the integrals gives

$$\begin{aligned}h_1(T) &= \frac{i}{T}\left(\frac{1}{2}\int_0^T dt_1\int_0^T dt_2\,\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) - \int_0^T dt_1\int_0^{t_1} dt_2\,\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\right)\\ &= \frac{i}{T}\left(\frac{1}{2}\iint_{\mathcal{R}} dt_1dt_2\,\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) + \frac{1}{2}\iint_{\tilde{\mathcal{R}}} dt_1dt_2\,\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) - \int_{\mathcal{R}} dt_1dt_2\,\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\right),\end{aligned}$$

  where $\mathcal{R}\equiv\{(t_1,t_2)|0\le t_2\le t_1\le T\}$ and $\tilde{\mathcal{R}}\equiv\{(t_1,t_2)|0\le t_1\le t_2\le T\}$ partition the square.



  Note that exchanging $t_1\leftrightarrow t_2$ gives $\mathcal{R}\leftrightarrow\tilde{\mathcal{R}}$. This finally gives

$$\boxed{h_1(T) = \frac{-i}{2T}\int_0^T dt_1\int_0^{t_1} dt_2\,[\delta\tilde{H}(t_1),\delta\tilde{H}(t_2)]}$$

- *Second-order term:* For $m = 2$, the configurations are (3), (1, 2), (2, 1) and (1, 1, 1). Thus

$$h_2(T) = -\frac{1}{T}\Big[\mathcal{D}_3(\delta\tilde{H}, T) - \frac{1}{2}\big(\mathcal{D}_1(\delta\tilde{H}, T)\mathcal{D}_2(\delta\tilde{H}, T) + \mathcal{D}_2(\delta\tilde{H}, T)\mathcal{D}_1(\delta\tilde{H}, T)\big) + \frac{1}{3}\mathcal{D}_1^3(\delta\tilde{H}, T)\Big].$$

Now write out the integrals:

$$\begin{aligned} Th_2(T) = &-\int_0^T dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3\, \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) \\ &- \frac{1}{3}\int_0^T dt_1 \int_0^T dt_2 \int_0^T dt_3\, \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) \\ &+ \frac{1}{2}\int_0^T dt_1 \int_0^T dt_2 \int_0^{t_2} dt_3 \Big(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\Big). \end{aligned}$$

Write the three contributions as

$$\begin{aligned} Th_2(T) = &-\underbrace{\int_{\mathcal{R}} d^3t\, \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3)}_{I_1} - \frac{1}{3}\underbrace{\int_{[0,T]^3} d^3t\, \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3)}_{I_2} \\ &+ \frac{1}{2}\underbrace{\int_{\mathcal{R}'} d^3t \left(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\right)}_{I_3}. \end{aligned} \tag{2}$$

where the integration domains are $\mathcal{R} = \mathcal{R}_{123}$ and $\mathcal{R}' = \mathcal{R}_{123} \cup \mathcal{R}_{213} \cup \mathcal{R}_{312}$, where we define the shorthand

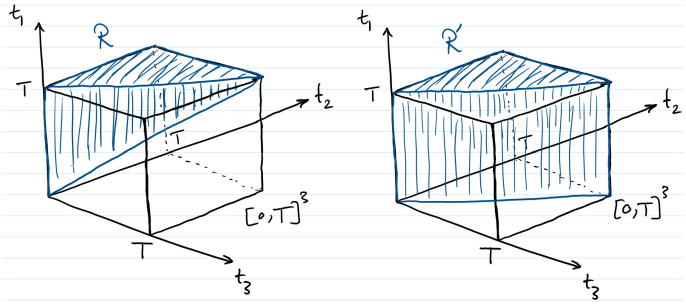
$$\mathcal{R}_\pi \equiv \{(t_{\pi(1)}, t_{\pi(2)}, t_{\pi(3)}) | 0 \le t_3 \le t_2 \le t_1 \le T\},$$

for any given permutation $\pi \in S_3$. The regions $\{\mathcal{R}_\pi\}_{\pi\in S_3}$ partition the cube:

$$\begin{cases} [0,T]^3 = \bigcup\limits_{\pi\in S_3} \mathcal{R}_\pi & \text{(union over all permutations)} \\ \mathcal{R}_\pi \cap \mathcal{R}_{\pi'} = \emptyset & \pi \neq \pi' \text{ (up to boundary points unimportant for integrals)} \end{cases} \tag{3}$$

We can see this explicitly by realizing that:

$$\begin{aligned} \mathcal{R}_{123} &= \{(t_1, t_2, t_3) \mid 0 \le t_3 \le t_2 \le t_1 \le T\} \\ \mathcal{R}_{213} &= \{(t_1, t_2, t_3) \mid 0 \le t_3 \le t_1 \le t_2 \le T\} \\ \mathcal{R}_{312} &= \{(t_1, t_2, t_3) \mid 0 \le t_1 \le t_3 \le t_2 \le T\} \end{aligned}$$



$$\mathcal{R}_{123} \cup \mathcal{R}_{213} \cup \mathcal{R}_{312} = \{(t_1, t_2, t_3) \mid 0 \le t_1 \le T, \quad 0 \le t_3 \le t_2 \le T\} = \mathcal{R}' \quad \checkmark$$

With that in mind, the second contribution is

$$\begin{aligned} I_2 &= \int_{[0,T]^3} d^3t\, \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) = \sum_\pi \int_{\mathcal{R}_\pi} d^3t\, \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) \\ &= \int_{\mathcal{R}} d^3t \Big(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_1)\delta\tilde{H}(t_3)\delta\tilde{H}(t_2) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_1)\delta\tilde{H}(t_3) \\ &\qquad + \delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) + \delta\tilde{H}(t_3)\delta\tilde{H}(t_2)\delta\tilde{H}(t_1) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\Big) \end{aligned} \tag{4}$$

For the third contribution, use Eq. (3) and make the changes of variables $(t_1, t_2, t_3) = (\tilde{t}_2, \tilde{t}_1, \tilde{t}_3)$ in $\mathcal{R}_{213}$ and $(t_1, t_2, t_3) = (\tilde{t}_3, \tilde{t}_1, \tilde{t}_2)$ in $\mathcal{R}_{312}$. This gives

$$\begin{aligned}
I_3 &= \int_{\mathcal{R}} d^3t \left(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\right) \\
&\quad + \int_{\mathcal{R}_{213}} d^3t \left(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\right) \\
&\quad + \int_{\mathcal{R}_{312}} d^3t \left(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\right) \\
&= \int_{\mathcal{R}} d^3t \Big(\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1) \\
&\qquad + \delta\tilde{H}(t_2)\delta\tilde{H}(t_1)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_1)\delta\tilde{H}(t_3)\delta\tilde{H}(t_2) \\
&\qquad + \delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) + \delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\Big) \\
&= \int_{\mathcal{R}} d^3t \Big(2\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1) \\
&\qquad + \delta\tilde{H}(t_2)\delta\tilde{H}(t_1)\delta\tilde{H}(t_3) + \delta\tilde{H}(t_1)\delta\tilde{H}(t_3)\delta\tilde{H}(t_2) + \delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\Big). \qquad (5)
\end{aligned}$$

Combining Eqs. (2)–(5) gives

$$\begin{aligned}
Th_2(T) &= \int_{\mathcal{R}} d^3t \Big[\left(-1 - \frac{1}{3} + 1\right)\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) \\
&\quad - \frac{1}{3}\delta\tilde{H}(t_3)\delta\tilde{H}(t_2)\delta\tilde{H}(t_1) \\
&\quad + \left(-\frac{1}{3} + \frac{1}{2}\right)\Big(\delta\tilde{H}(t_1)\delta\tilde{H}(t_3)\delta\tilde{H}(t_2) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_1)\delta\tilde{H}(t_3) \\
&\qquad + \delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) + \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\Big)\Big].
\end{aligned}$$

$$\begin{aligned}
h_2(T) &= \frac{-1}{6T}\int_{\mathcal{R}} d^3t \Big[2\delta\tilde{H}(t_1)\delta\tilde{H}(t_2)\delta\tilde{H}(t_3) + 2\delta\tilde{H}(t_3)\delta\tilde{H}(t_2)\delta\tilde{H}(t_1) \\
&\quad - \delta\tilde{H}(t_1)\delta\tilde{H}(t_3)\delta\tilde{H}(t_2) - \delta\tilde{H}(t_2)\delta\tilde{H}(t_1)\delta\tilde{H}(t_3) \\
&\quad - \delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\delta\tilde{H}(t_2) - \delta\tilde{H}(t_2)\delta\tilde{H}(t_3)\delta\tilde{H}(t_1)\Big]
\end{aligned}$$

Finally, this gives

$$\boxed{h_2(T) = \frac{-1}{6T}\int_0^T dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \Big([\delta\tilde{H}(t_1), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_1)]]\Big)}$$

as desired.

Summary of First Few Terms

Setting $\lambda = 1$, the first few terms in the expansion are

$$\begin{aligned}
-ih_0(T) &= \frac{1}{T}\int_0^T dt\,(-i\delta\tilde{H}(t)) \\
-ih_1(T) &= \frac{1}{2T}\int_0^T dt_1 \int_0^{t_1} dt_2\,[-i\delta\tilde{H}(t_1), -i\delta\tilde{H}(t_2)] \\
-ih_2(T) &= \frac{1}{6T}\int_0^T dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \Big(\underbrace{[-i\delta\tilde{H}(t_1), [-i\delta\tilde{H}(t_2), -i\delta\tilde{H}(t_3)]] + [-i\delta\tilde{H}(t_3), [-i\delta\tilde{H}(t_2), -i\delta\tilde{H}(t_1)]]}_{\text{Note } t_3 \leftrightarrow t_1 \text{ symmetry!}}\Big)
\end{aligned}$$

which gives the interaction-frame propagator $\tilde{U}(T) = e^{-iT\sum_{m=0}^{\infty} h_m(T)}$, and consequently the original-frame propagator $U(T) = U_0(T)\tilde{U}(T)$.

4 Useful Properties of Average Hamiltonians

2. *Lemma:* We call the unperturbed evolution *cyclic* when it is T-periodic, i.e. $U_0(t+T) = U_0(t)$ (up to a global phase). In this case, the original frame and interaction frame coincide at $t = nT$, since $U_0(nT) = U_0(0) = \mathbb{1}$, meaning that

$$U(nT) = U_0(nT)\tilde{U}(nT) = \tilde{U}(nT) \qquad n \in \mathbb{Z}$$

Additionally, if the perturbation Hamiltonian is also T-periodic, we have $\tilde{U}(nT) = \tilde{U}^n(T) = e^{-inT\bar{H}(T)}$, i.e.

$$U(nT) = e^{-inT\bar{H}(T)} \qquad n \in \mathbb{Z}$$

So as long as the system is observed stroboscopically at $t = nT$, the dynamics in the original frame is described by only the interaction-frame average Hamiltonian.

3. *Lemma:* Notice how our ansatz $\bar{H}(T) = \sum_{m=0}^{\infty} \lambda^{m+1} h_m(T)$ starts with an $\mathcal{O}(\lambda)$ term. With this condition, the average Hamiltonian is uniquely determined as a power series in λ. However, had we started from a $\mathcal{O}(\lambda^0)$ term, it would not have been unique, e.g. for any $k \in \mathbb{Z}$

$$\bar{H}'(T) \equiv \frac{2\pi k}{T}\mathbb{1} + \bar{H}(T) \Rightarrow e^{-iT\bar{H}'(T)} = e^{-iT\bar{H}(T)},$$

which indicates two different average Hamiltonians for the same propagator.

4. *Lemma:* All Magnus terms are Hermitian. Hence truncating the average Hamiltonian at any order gives a unitary approximation to the propagator.

Proof: Since the exact propagator $\tilde{U}(T)$ is unitary, $\tilde{U}^\dagger(T) = \tilde{U}^{-1}(T)$, resulting in $e^{iT\bar{H}^\dagger(T)} = e^{iT\bar{H}(T)}$. Both $\bar{H}(T)$ and $\bar{H}^\dagger(T)$ start at $\mathcal{O}(\lambda)$. By Lemma 3, the power series in the exponent is uniquely determined. Therefore, $\bar{H}^\dagger(T) = \bar{H}(T)$. Equating powers of λ gives $h_m^\dagger(T) = h_m(T)$.

5. *Lemma:* The mth Magnus term is order $\|h_m(T)\| = \mathcal{O}\left(T^m \|\delta H\|_{\max}^{m+1}\right)$, where the implicit constant only depends on m.

Proof: Using the Dyson-to-Magnus conversion in Eq. (1), the triangle inequality and submultiplicativity give

$$\|h_m(T)\| \leq \frac{1}{T}\sum_{k=1}^{m+1}\frac{1}{k}\sum_{\substack{n_1+\cdots+n_k=m+1 \\ n_1,\ldots,n_k\in\mathbb{N}}}\prod_{l=1}^{k}\left\|\mathcal{D}_{n_l}(\delta\tilde{H},T)\right\|.$$

The Dyson bound $\|\mathcal{D}_{n_l}(\delta\tilde{H},T)\| \leq (T\|\delta H\|_{\max})^{n_l}/n_l!$ then gives

$$\|h_m(T)\| \leq T^m\|\delta H\|_{\max}^{m+1}\sum_{k=1}^{m+1}\frac{1}{k}\sum_{\substack{n_1+\cdots+n_k=m+1 \\ n_1,\ldots,n_k\in\mathbb{N}}}\frac{1}{n_1!\cdots n_k!} \leq T^m\|\delta H\|_{\max}^{m+1}\sum_{k=1}^{m+1}\frac{1}{k}\binom{m}{k-1} = \mathcal{O}\left(T^m\|\delta H\|_{\max}^{m+1}\right).$$

This gives us a physical criterion for when truncating the Magnus series is a good approximation. Since $Th_m(T) = \mathcal{O}\left(T^{m+1}\|\delta H\|_{\max}^{m+1}\right)$ appears in the exponent of the propagator, we require that the time T has to be very short compared to the time-scale set by the perturbation Hamiltonian. The next lemma makes this criterion more precise.

6. *Lemma:* Let $\bar{H}^{(\leq m)}(T) = \sum_{k=0}^{m} h_k(T)$ be the truncated Magnus series up to the mth term. Define $\Delta\bar{H} \equiv \bar{H} - \bar{H}^{(\leq m)}$ as the part of the average Hamiltonian that is thrown out by the truncation. The corresponding error in the propagator is bounded as:

$$\|e^{-iT\bar{H}^{(\leq m)}} - e^{-iT\bar{H}}\| \leq T\|\Delta\bar{H}\|.$$

Proof: Define the auxiliary function $F(x) \equiv e^{-iT(\bar{H}^{(\leq m)}+x\Delta\bar{H})}$, such that $F(0) = e^{-iT\bar{H}^{(\leq m)}}$ and $F(1) = e^{-iT\bar{H}}$. Using the derivative of the matrix exponential (Appendix D), we have

$$F'(x) = \int_0^1 ds\, e^{-iT(1-s)(\bar{H}^{(\leq m)}+x\Delta\bar{H})}(-iT\Delta\bar{H})e^{-iTs(\bar{H}^{(\leq m)}+x\Delta\bar{H})}$$

$$\Rightarrow \|F'(x)\| \leq T \int_0^1 ds\, \|\Delta \bar{H}\| = T\|\Delta \bar{H}\| \qquad \text{(unitary invariance, } \triangle \text{ inequality)} \tag{6}$$

Now,

$$e^{-iT\bar{H}^{(\leq m)}} - e^{-iT\bar{H}} = F(0) - F(1) = -\int_0^1 dx\, F'(x)$$

$$\Rightarrow \|e^{-iT\bar{H}^{(\leq m)}} - e^{-iT\bar{H}}\| \leq \int_0^1 dx\, \|F'(x)\| \qquad (\triangle \text{ inequality})$$

Using Eq. (6), we get the final result

$$\boxed{\|e^{-iT\bar{H}^{(\leq m)}} - e^{-iT\bar{H}}\| \leq T\|\Delta \bar{H}\|.}$$

Note that the bound holds for any unitarily invariant norm.

7. *Lemma:* For a symmetric interaction-frame Hamiltonian $\delta\tilde{H}(t) = \delta\tilde{H}(T-t)$, all odd-order Magnus terms vanish, i.e. $h_m(T) = 0$ for all odd m.

   Proof: First see what the symmetry implies for the Dyson terms by changing integration variables as $\tilde{t}_1 = T - t_n, \ldots,$ and $\tilde{t}_n = T - t_1$, under which the integration region

   $$\mathcal{R} = \{(t_1, \ldots, t_n) \mid 0 \leq t_n \leq \cdots \leq t_1 \leq T\}.$$

   is unchanged. This gives

   $$\begin{aligned}\mathcal{D}_n(\delta\tilde{H}, T) &= \int_{\mathcal{R}} d\tilde{t}_1 \cdots d\tilde{t}_n\, \delta\tilde{H}(T-\tilde{t}_n)\cdots\delta\tilde{H}(T-\tilde{t}_1) \\ &= \int_{\mathcal{R}} dt_1 \cdots dt_n\, \delta\tilde{H}(t_n)\cdots\delta\tilde{H}(t_1) \\ &= \mathcal{D}_n^\dagger(\delta\tilde{H}, T).\end{aligned}$$

   Thus, for $\lambda \in \mathbb{R}$,

   $$\sum_{n=0}^{\infty}(-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H}, T) = \left(\sum_{n=0}^{\infty}(-i)^n(-\lambda)^n\mathcal{D}_n(\delta\tilde{H}, T)\right)^\dagger,$$

   i.e. the average Hamiltonian of the inverse propagator is equal to that of the forward propagator with the replacement $\lambda \to -\lambda$. Adding the λ dependence in the argument for clarity, we have $\tilde{U}^\dagger(T,\lambda) = \tilde{U}(T,-\lambda)$, i.e.

   $$e^{iT\bar{H}(T,\lambda)} = e^{-iT\bar{H}(T,-\lambda)}.$$

   The uniqueness in Lemma 3 gives $\bar{H}(T,\lambda) = -\bar{H}(T,-\lambda)$, and hence

   $$\sum_{m=0}^{\infty}\lambda^{m+1}\left[1+(-1)^{m+1}\right]h_m(T) = 0,$$

   which results in $h_m(T) = 0$ for odd m, as desired.

8. *Lemma:* For an anti-symmetric interaction-frame Hamiltonian $\delta\tilde{H}(t) = -\delta\tilde{H}(T-t)$, all Magnus terms vanish, i.e. $\bar{H}(T) = 0$ and $\tilde{U}(T) = \mathbb{1}$.

   Proof: The same substitution as the previous lemma gives

   $$\begin{aligned}\mathcal{D}_n(\delta\tilde{H}, T) &= \int_{\mathcal{R}} d\tilde{t}_1 \cdots d\tilde{t}_n\, \delta\tilde{H}(T-\tilde{t}_n)\cdots\delta\tilde{H}(T-\tilde{t}_1) \\ &= (-1)^n \int_{\mathcal{R}} dt_1 \cdots dt_n\, \delta\tilde{H}(t_n)\cdots\delta\tilde{H}(t_1) \\ &= (-1)^n\mathcal{D}_n^\dagger(\delta\tilde{H}, T).\end{aligned}$$

Thus

$$\tilde{U}(T,\lambda) = \sum_{n=0}^{\infty} (-i)^n \lambda^n \mathcal{D}_n(\delta\tilde{H}, T) = \tilde{U}^\dagger(T,\lambda),$$

which results in $e^{-iT\bar{H}(T,\lambda)} = e^{iT\bar{H}(T,\lambda)}$, and consequently $\bar{H}(T,\lambda) = -\bar{H}(T,\lambda) = 0$. Thus $h_m(T) = 0$ for every $m \in \mathbb{W}$. Therefore, $\bar{H}(T) = 0$ and $\tilde{U}(T) = \mathbb{1}$.

9. *Lemma:* If $h_j(T) = 0$ for $j \in \{0, 1, \ldots, m-1\}$, the mth term simplifies to:

$$h_m(T) = \frac{(-i)^m}{T} \int_0^T dt_1 \cdots \int_0^{t_m} dt_{m+1}\, \delta\tilde{H}(t_1) \cdots \delta\tilde{H}(t_{m+1}).$$

Proof:

$$e^{-iT\sum_{j=0}^{\infty} \lambda^{j+1} h_j(T)} = e^{-iT\sum_{j=m}^{\infty} \lambda^{j+1} h_j(T)} = \sum_{n=0}^{\infty} (-i)^n \lambda^n \mathcal{D}_n(\delta\tilde{H}, T)$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{(-iT)^k}{k!} \left(\sum_{j=m}^{\infty} \lambda^{j+1} h_j(T) \right)^k = \sum_{n=0}^{\infty} (-i)^n \lambda^n \mathcal{D}_n(\delta\tilde{H}, T)$$

Equating powers of λ^{m+1} gives

$$-iT h_m(T) = (-i)^{m+1} \mathcal{D}_{m+1}(\delta\tilde{H}, T)$$

and hence

$$\boxed{h_m(T) = \frac{(-i)^m}{T} \int_0^T dt_1 \cdots \int_0^{t_m} dt_{m+1}\, \delta\tilde{H}(t_1) \cdots \delta\tilde{H}(t_{m+1})}$$

10. *Lemma:* For a symmetric $\delta\tilde{H}$, the 0th order term can be calculated from just half the evolution:

$$\boxed{h_0(T) = \frac{1}{T} \int_0^T dt\, \delta\tilde{H}(t) = \frac{1}{T} \int_0^{T/2} dt\, (\delta\tilde{H}(t) + \delta\tilde{H}(T-t)) = \frac{1}{T/2} \int_0^{T/2} dt\, \delta\tilde{H}(t)}$$

11. *Lemma:* For a symmetric $\delta\tilde{H}$, the second-order Magnus term simplifies as follows:

$$h_2(T) = \frac{-1}{3T} \left[\int_0^{T/2} dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \left([\delta\tilde{H}(t_1), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_1)]] \right) \right.$$
$$\left. + \int_{T/2}^{T} dt_1 \int_{T/2}^{t_1} dt_2 \int_0^{T/2} dt_3 \left([\delta\tilde{H}(t_1), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_1)]] \right) \right]$$

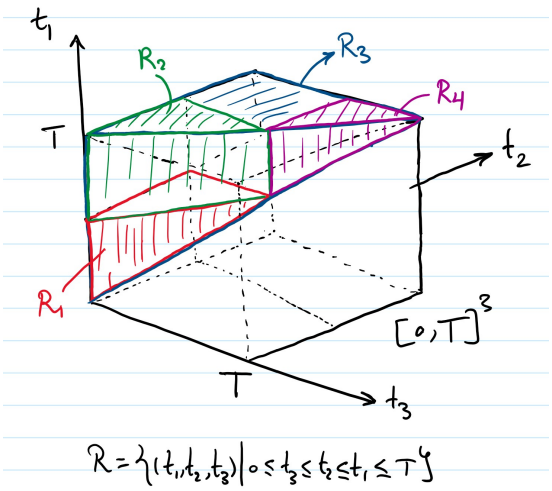
Furthermore, if $h_0(T) = 0$, we have the additional simplification:

$$\boxed{h_2(T) = \frac{-1}{3T} \int_0^{T/2} dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \left([\delta\tilde{H}(t_1), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_1)]] \right),}$$

which is an integral over the smaller region $\mathcal{R}_1$.

Proof: Partition the integration region $\mathcal{R}$ into 4 subregions

$$\mathcal{R} = \bigcup_{i=1}^{4} \mathcal{R}_i, \quad \text{where} \quad \begin{cases} \mathcal{R}_1 = \{(t_1,t_2,t_3) \mid 0 \le t_3 \le t_2 \le t_1 \le T/2\} \\ \mathcal{R}_2 = \{(t_1,t_2,t_3) \mid 0 \le t_3 \le t_2 \le T/2,\ T/2 \le t_1 \le T\} \\ \mathcal{R}_3 = \{(t_1,t_2,t_3) \mid 0 \le t_3 \le T/2,\ T/2 \le t_2 \le t_1 \le T\} \\ \mathcal{R}_4 = \{(t_1,t_2,t_3) \mid T/2 \le t_3 \le t_2 \le t_1 \le T\} \end{cases}$$



Denote contribution of region $\mathcal{R}_i$ to $h_2(T)$ by I_i. Also use shorthand notation

$$f(t_1,t_2,t_3) = [\delta\tilde{H}(t_1),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_1)]] \quad \text{(integrand)}$$

$$I_4 = \frac{-1}{6T}\int_{T/2}^{T} dt_1 \int_{T/2}^{t_1} dt_2 \int_{T/2}^{t_2} dt_3\, f(t_1,t_2,t_3) = -\frac{1}{6T}\int_0^{T/2} d\tilde{t}_1 \int_0^{\tilde{t}_1} d\tilde{t}_2 \int_0^{\tilde{t}_2} d\tilde{t}_3\, f(T-\tilde{t}_3, T-\tilde{t}_2, T-\tilde{t}_1)$$

Change variables:

$$\begin{cases} \tilde{t}_1 = T - t_3 \\ \tilde{t}_2 = T - t_2 \\ \tilde{t}_3 = T - t_1 \end{cases} \qquad (T/2 \le t_3 \le t_2 \le t_1 \le T \iff 0 \le \tilde{t}_3 \le \tilde{t}_2 \le \tilde{t}_1 \le T/2)$$

Note that $f(t_1,t_2,t_3) = f(t_3,t_2,t_1)$, hence

$$\begin{aligned} I_4 &= -\frac{1}{6T}\int_0^{T/2} d\tilde{t}_1 \int_0^{\tilde{t}_1} d\tilde{t}_2 \int_0^{\tilde{t}_2} d\tilde{t}_3\, f(T-\tilde{t}_1, T-\tilde{t}_2, T-\tilde{t}_3) \\ &= -\frac{1}{6T}\int_0^{T/2} d\tilde{t}_1 \int_0^{\tilde{t}_1} d\tilde{t}_2 \int_0^{\tilde{t}_2} d\tilde{t}_3\, f(\tilde{t}_1,\tilde{t}_2,\tilde{t}_3) \qquad \text{(symmetry of } \delta\tilde{H}) \\ &= I_1 \end{aligned}$$

On the other hand,

$$\begin{aligned} I_2 &= -\frac{1}{6T}\int_{T/2}^{T} dt_1 \int_0^{T/2} dt_2 \int_0^{t_2} dt_3\, f(t_1,t_2,t_3) \\ &= \frac{-1}{6T}\int_{T/2}^{T} d\tilde{t}_1 \int_{T/2}^{\tilde{t}_1} d\tilde{t}_2 \int_0^{T/2} d\tilde{t}_3\, f(T-\tilde{t}_3, T-\tilde{t}_2, T-\tilde{t}_1) \end{aligned}$$

$$\begin{cases} \tilde{t}_1 = T - t_3 \\ \tilde{t}_2 = T - t_2 \\ \tilde{t}_3 = T - t_1 \end{cases} \qquad \begin{pmatrix} T/2 \le t_1 \le T, & 0 \le t_3 \le t_2 \le T/2 \\ & \Updownarrow \\ 0 \le \tilde{t}_3 \le T/2, & T/2 \le \tilde{t}_2 \le \tilde{t}_1 \le T \end{pmatrix}$$

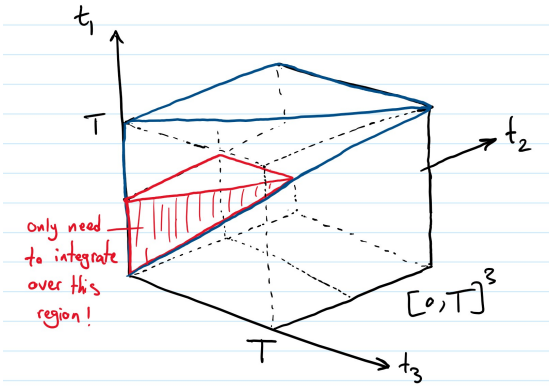
Using the symmetries of $\delta\tilde{H}$ and f,

$$I_2 = \frac{-1}{6T}\int_{T/2}^{T} d\tilde{t}_1 \int_{T/2}^{\tilde{t}_1} d\tilde{t}_2 \int_0^{T/2} d\tilde{t}_3\, f(\tilde{t}_1,\tilde{t}_2,\tilde{t}_3) = I_3$$

$$h_2(T) = \frac{-1}{6T}\Bigg[2\int_0^{T/2} dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \Big([\delta\tilde{H}(t_1),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_1)]]\Big)$$
$$+ 2\int_{T/2}^{T} dt_1 \int_{T/2}^{t_1} dt_2 \int_0^{T/2} dt_3 \Big([\delta\tilde{H}(t_1),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_1)]]\Big)\Bigg]$$

Now if $h_0(T) \propto \int_0^T dt\, \delta\tilde{H}(t) \propto \int_0^{T/2} dt\, \delta\tilde{H}(t) = 0$, the second integral above vanishes, since $\int_0^{T/2} dt_3$ can be separated. Thus

$$\boxed{h_2(T) = \frac{-1}{3T}\int_0^{T/2} dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \Big([\delta\tilde{H}(t_1),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3),[\delta\tilde{H}(t_2),\delta\tilde{H}(t_1)]]\Big)} \quad \checkmark$$



12. *Lemma:* Magnus-to-Dyson conversion. For $n \in \mathbb{N}$,

$$\mathcal{D}_n(\delta\tilde{H},T) = \sum_{k=1}^{n}(-i)^{k-n}\frac{T^k}{k!}\sum_{\substack{m_1+\cdots+m_k=n-k \\ m_1,\ldots,m_k\in\mathbb{W}}}\prod_{l=1}^{k} h_{m_l}(T).$$

The zeroth Dyson term is still $\mathcal{D}_0(\delta\tilde{H},T) = \mathbb{1}$.

Proof: The interaction-frame propagator is

$$\tilde{U}(T,\lambda) = e^{-iT\sum_{m=0}^{\infty}\lambda^{m+1}h_m(T)} = \mathbb{1} + \sum_{n=1}^{\infty}(-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H},T).$$

The coefficient of λ^{m+1} in $\bar{H}(T,\lambda)$ is $h_m(T)$. Now expand the exponential:

$$\sum_{k=0}^{\infty}\frac{(-iT)^k}{k!}\left(\sum_{m=0}^{\infty}\lambda^{m+1}h_m(T)\right)^k = \mathbb{1} + \sum_{n=1}^{\infty}(-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H},T).$$

Expanding the products gives

$$\sum_{k=1}^{\infty}\sum_{m_1=0}^{\infty}\cdots\sum_{m_k=0}^{\infty}\frac{(-iT)^k}{k!}\lambda^{k+\sum_{l=1}^{k}m_l}\prod_{l=1}^{k}h_{m_l}(T) = \sum_{n=1}^{\infty}(-i)^n\lambda^n\mathcal{D}_n(\delta\tilde{H},T).$$

Equating powers of λ^n, we have

$$(-i)^n\mathcal{D}_n(\delta\tilde{H},T) = \sum_{k=1}^{n}\frac{(-iT)^k}{k!}\sum_{\substack{m_1+\cdots+m_k=n-k \\ m_1,\ldots,m_k\in\mathbb{W}}}\prod_{l=1}^{k}h_{m_l}(T).$$

Thus,

$$\boxed{\mathcal{D}_n(\delta\tilde{H},T)=\sum_{k=1}^{n}(-i)^{k-n}\frac{T^k}{k!}\sum_{\substack{m_1+\cdots+m_k=n-k\\ m_1,\ldots,m_k\in\mathbb{W}}}\prod_{l=1}^{k}h_{m_l}(T)}. \tag{7}$$

5 Symmetry of Control Sequences

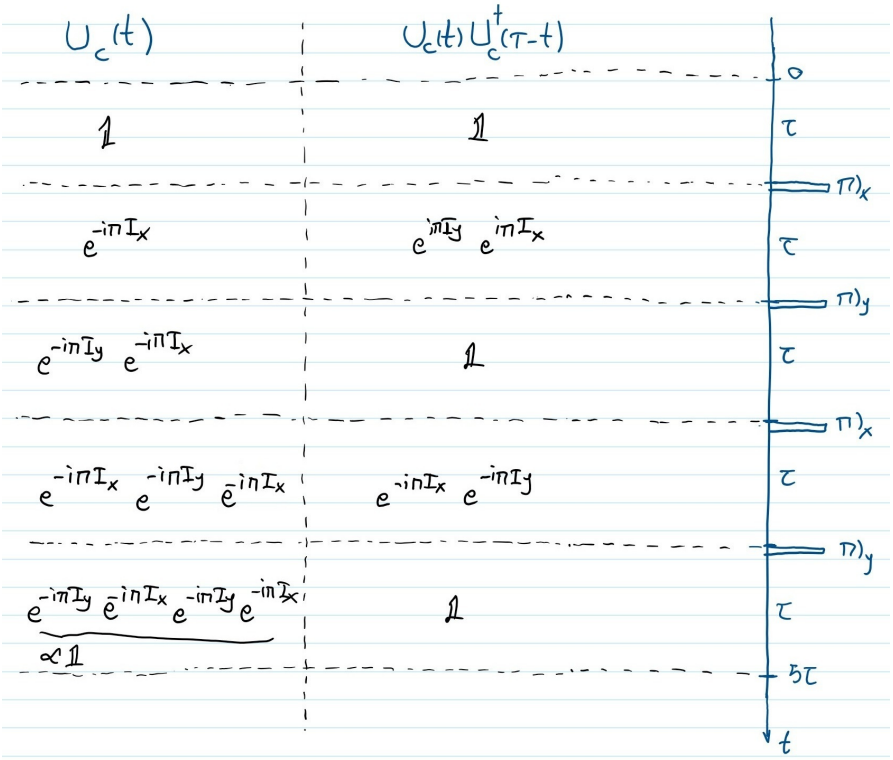
- Here we are concerned with the typical case where $H_0(t) = H_C(t)$ is the control Hamiltonian and δH is time-independent. Denote the control unitary by $U_C(t)$. We start in the rotating frame. The interaction frame of $H_C(t)$ is the toggling frame.

5.1 Symmetric Sequences

- We want $\delta\tilde{H}(t) = \delta\tilde{H}(T-t)$ to make all odd Magnus terms vanish. This requires

$$U_C^\dagger(t)\,\delta H\,U_C(t) = U_C^\dagger(T-t)\,\delta H\,U_C(T-t) \Rightarrow \boxed{[\delta H, U_C(t)U_C^\dagger(T-t)]=0}$$

- *Example:* consider the following sequence [1] in the presence of a resonance offset $\delta H \propto I_z$:



where we have dropped global phases. Thus

$$[\delta H, U_C(t)U_C^\dagger(T-t)]=0.$$

Hence, the toggling-frame Hamiltonian is symmetric. Note that the given sequence only gives a symmetric $\delta\tilde{H}$ for the *specific* perturbation given. Also note that this sequence is cyclic up to an overall phase.

- Do sequences exist that give a symmetric $\delta\tilde{H}$ for an arbitrary (time-independent) δH? Yes!

$$[\delta H, U_C(t)U_C^\dagger(T-t)]=0 \quad \forall \delta H$$

[1] Not to be confused with an XYXY! It actually rescales resonance offsets by 1/5 to leading order.

This gives

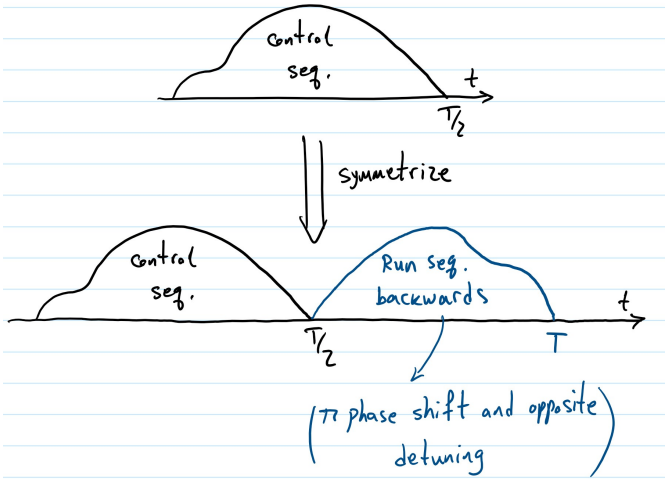
$$U_C(t)U_C^\dagger(T-t) = \mathbb{1} \qquad \text{(up to a phase)}.$$

So

$$\boxed{U_C(t) = U_C(T-t) \qquad \text{(up to a phase)}}$$

would generate a symmetric $\delta\tilde{H}$ for arbitrary δH.

- We can use this to symmetrize a sequence in practice:



Apply the time-reversed control with opposite sign in the second half:

$$H_C(t) = \mathbf{b}(t)\cdot\mathbf{I}, \qquad H_C(T-t) = -H_C(t), \qquad \mathbf{b}(t) = \omega_{1x}(t)\hat{\mathbf{x}} + \omega_{1y}(t)\hat{\mathbf{y}} + \Delta\omega(t)\hat{\mathbf{z}}.$$

This requires a π phase shift and opposite detuning to run the evolution backwards.

Proof: The control unitary satisfies

$$i\dot{U}_C(t) = H_C(t)U_C(t), \qquad U_C(0) = \mathbb{1}.$$

Define $V(t) = U_C(T-t)$ for $T/2 \le t \le T$. Then

$$i\dot{V}(t) = -i\dot{U}_C(T-t) = -H_C(T-t)U_C(T-t) = H_C(t)V(t).$$

Thus $V(t)$ and $U_C(t)$ satisfy the same differential equation on the second half, with the same midpoint value $V(T/2) = U_C(T/2)$. Uniqueness gives

$$\boxed{U_C(T-t) = U_C(t)}$$

on the second half, and hence on the whole interval. For any static δH,

$$\delta\tilde{H}(T-t) = U_C^\dagger(T-t)\delta H U_C(T-t) = U_C^\dagger(t)\delta H U_C(t) = \delta\tilde{H}(t).$$

Setting $t = 0$ also gives $U_C(T) = \mathbb{1}$. These are exact equalities for this construction.

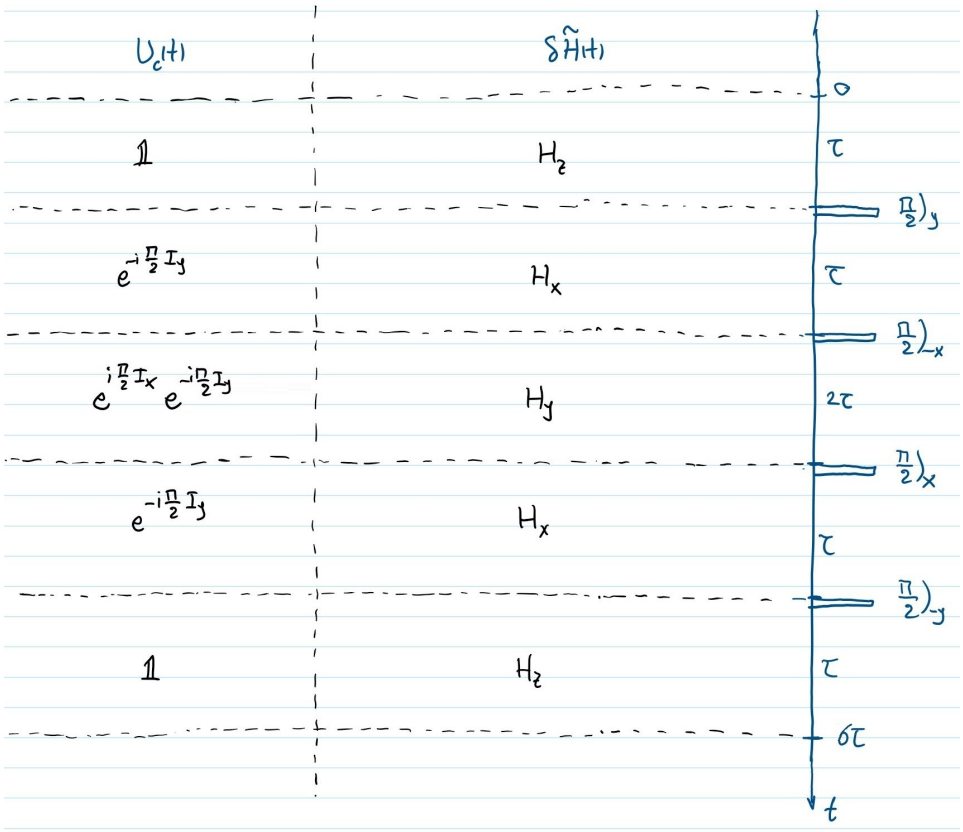
Summary of Symmetrized Sequences

1. $\delta\tilde{H}(T-t) = \delta\tilde{H}(t)$ for any static δH
2. Sequence automatically satisfies $U_C(T) = \mathbb{1}$, and for repeated applications, it is cyclic with stroboscopic rotating-frame propagator $U(nT) = e^{-inT\tilde{H}(T)}$.
3. AHT simplifies:
   - $h_0(T)$ is the average of half the sequence.
   - All $h_m(T) = 0$, for odd m.
   - $h_2(T)$ is simplified. Especially if $h_0 = 0$, we only need an integral over $0 \le t_3 \le t_2 \le t_1 \le T/2$.

- *Example:* Consider the WAHUHA sequence below under the secular homonuclear dipolar Hamiltonian $\delta H = H_z$, where

$$H_\alpha = \sum_{j<k} d_{jk}(3I_j^\alpha I_k^\alpha - \mathbf{I}_j \cdot \mathbf{I}_k), \qquad \alpha \in \{x, y, z\}.$$

Notice that $H_x + H_y + H_z = 0$.



For ideal δ-pulses, the WAHUHA sequence gives

$$h_0(6\tau) = \frac{2\tau}{6\tau}(H_x + H_y + H_z) = 0.$$

Furthermore, the toggling-frame Hamiltonian is symmetric in time, so all odd-order Magnus terms vanish by Lemma 7.

However, $h_2(6\tau)$ is generally nonzero. Since the toggling-frame Hamiltonian is symmetric and $h_0(6\tau) = 0$, we can use Lemma 11 to calculate it from just the first half of the sequence:

$$\begin{aligned}
h_2(6\tau) &= -\frac{1}{18\tau}\int_0^{3\tau} dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \left([\delta\tilde{H}(t_1), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_3)]] + [\delta\tilde{H}(t_3), [\delta\tilde{H}(t_2), \delta\tilde{H}(t_1)]]\right) \\
&= -\frac{\tau^2}{18}\left(\frac{1}{2}\Big([H_z - H_x, [H_z, H_x]] + [H_x - H_y, [H_x, H_y]] + [H_y - H_z, [H_y, H_z]]\Big) + [H_y, [H_x, H_z]] + [H_z, [H_x, H_y]]\right) \\
&= -\frac{\tau^2}{18}\Big([H_y, [H_x, H_z]] + [H_z, [H_x, H_y]]\Big) \\
&= \boxed{-\frac{\tau^2}{18}[H_z - H_y, [H_z, H_x]]}, \qquad (8)
\end{aligned}$$

where we used $H_y = -H_x - H_z$, which gives $[H_x, H_y] = -[H_x, H_z] = [H_z, H_x]$ and $[H_y, H_z] = -[H_x, H_z] = [H_z, H_x]$. Thus, WAHUHA averages out the dipolar Hamiltonian up to first order, but leaves a generally nonzero second-order term.

5.2 Anti-Symmetric Sequences?

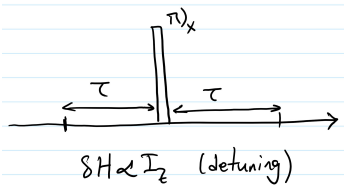
- What do we need to get $\delta\tilde{H}(T-t) = -\delta\tilde{H}(t)$, which makes the average Hamiltonian vanish?

$$U_C^\dagger(T-t)\,\delta H\,U_C(T-t) = -U_C^\dagger(t)\,\delta H\,U_C(t)$$

$$\Rightarrow \underbrace{\{\delta H, U_C(T-t)U_C^\dagger(t)\}}_{\text{anti-commutator}} = 0$$

- Unlike the symmetric case, there is no sequence that makes $\delta\tilde{H}$ anti-symmetric for an arbitrary δH. This is because the only operator that anticommutes with everything is the zero operator[1].

- What about specific perturbations? One example is a Hahn echo with δ-pulses for a static resonance offset $\delta H = \Delta\omega I_z$. The π pulse is instantaneous, so the toggling-frame Hamiltonian is anti-symmetric at all times other than exactly $t = \tau$, which is just an isolated point that does not affect the integrals.



$$U_C(t) = \begin{cases} \mathbb{1} & 0 \le t < \tau \\ e^{-i\pi I_x} & \tau \le t \le 2\tau \end{cases} \Rightarrow \delta\tilde{H}(t) = \begin{cases} \Delta\omega I_z & 0 \le t < \tau \\ -\Delta\omega I_z & \tau \le t \le 2\tau \end{cases}$$

Thus,

$$\delta\tilde{H}(T-t) = -\delta\tilde{H}(t) \Rightarrow \boxed{\bar{H}(2\tau) = 0}.$$

The resonance offset is averaged out to all orders, for any $\Delta\omega$. Here, truncation is exact even when $T\|\delta H\|_{\max}$ is arbitrarily large (as long as the pulse is instantaneous). This shows that our truncation condition is sufficient to get a good approximation to the propagator, but it is not necessary.

- A different way to see this:

$$[\delta\tilde{H}(t_1), \delta\tilde{H}(t_2)] = 0 \qquad \forall t_1, t_2$$

Thus, $\tilde{U}(2\tau) = \text{Texp}\left[-i\int_0^{2\tau} dt\ \delta\tilde{H}(t)\right] = e^{-i\int_0^{2\tau} dt\ \delta\tilde{H}(t)}$, meaning that

$$\boxed{\bar{H}(2\tau) = \frac{1}{2\tau}\int_0^{2\tau} dt\,\delta\tilde{H}(t) = 0}$$

- *Claim:* Exact anti-symmetry with finite-amplitude pulses is impossible for a static non-zero δH.

  Proof: Suppose that $\delta\tilde{H}(T-t) = -\delta\tilde{H}(t)$. Finite-amplitude pulses give a finite $H_C(t)$, and thus continuous $U_C(t)$. The toggling-frame Hamiltonian is therefore also continuous. Anti-symmetry then requires $\delta\tilde{H}(T/2) = -\delta\tilde{H}(T/2)$, which implies $\delta\tilde{H}(T/2) = U_C^\dagger(T/2)\delta H U_C(T/2) = 0$. Multiplying by $U_C(T/2)$ and $U_C^\dagger(T/2)$ from the left and the right, respectively, results in $\delta H = 0$, i.e. the trivial case of no perturbation. So for any non-zero static δH and finite control amplitudes, anti-symmetry in the toggling-frame Hamiltonian is impossible.

  The δ-pulse Hahn echo above avoids this contradiction because $U_C(t)$ is discontinuous at $t = \tau$ due to the infinite control amplitude. So anti-symmetry holds everywhere except at that single point, which does not affect the integrals.

[1] If you are not convinced, try setting its anticommutator with the identity to zero.

6 Average Hamiltonians of Supersequences

- Suppose we construct a longer sequence (a *supersequence*) by concatenating M shorter subsequences, with durations $\tau_1, \ldots, \tau_M$ and total duration $T = \sum_{j=1}^{M} \tau_j$. Define $t_0 = 0$ and $t_j = \sum_{k=1}^{j} \tau_k$ as the time one sequence switches to the next.

  The interaction-frame propagator of the jth subsequence is

  $$\tilde{U}_j(\tau_j, \lambda) = \mathrm{Texp}\left[-i\lambda \int_{t_{j-1}}^{t_j} dt\, \delta\tilde{H}(t)\right] = e^{-i\tau_j \bar{H}_j(\tau_j,\lambda)},$$

  where

  $$\bar{H}_j(\tau_j, \lambda) = \sum_{m=0}^{\infty} \lambda^{m+1} h_m^{(j)}(\tau_j).$$

  To avoid clutter, we will omit the time arguments of the individual Magnus terms below.

- The propagator of the supersequence is:

  $$\tilde{U}(T, \lambda) = \prod_{j=1}^{M} \tilde{U}_j(\tau_j, \lambda) = \prod_{j=1}^{M} e^{-i\tau_j \sum_{m=0}^{\infty} \lambda^{m+1} h_m^{(j)}}.$$

- To derive the average Hamiltonian of the supersequence $\bar{H}$, we can simply use Eq. 10 from Appendix C to combine the product of exponentials into one exponent, and then equate the result to $e^{-iT\bar{H}} = e^{-iT\sum_{m=0}^{\infty} \lambda^{m+1} h_m}$ one λ power at a time.

- The first two terms are

  $$\boxed{\begin{aligned} h_0(T) &= \frac{1}{T}\sum_{j=1}^{M} \tau_j h_0^{(j)} \\ h_1(T) &= \frac{1}{T}\sum_{j=1}^{M} \tau_j h_1^{(j)} - \frac{i}{2T}\sum_{j>k} \tau_j \tau_k [h_0^{(j)}, h_0^{(k)}] \end{aligned}}$$

- *Important:* All subsequence average Hamiltonians $\bar{H}^{(j)}$ above must be expressed in the same interaction frame. If a subsequence's Magnus terms are calculated in a local toggling frame starting at the beginning of that subsequence, they must first be transformed to a global toggling frame as $h_m^{(j)} \to U_C^\dagger(t_{j-1}) h_m^{(j)} U_C(t_{j-1})$. Of course, if every subsequence is cyclic, i.e. $U_C(t_j) = \mathbb{1}$, then this would be no problem since the toggling frame would coincide with the rotating frame at the end of each subsequence.

6.1 Pulse-Cycle Decoupling

- The above results become particularly useful when some of the lower-order Magnus terms vanish within each subsequence. This is the basis of *pulse-cycle decoupling*.

- *Claim:* Suppose the first p Magnus terms vanish for every subsequence:

  $$h_0^{(j)} = h_1^{(j)} = \cdots = h_{p-1}^{(j)} = 0, \qquad j \in \{1, \ldots, M\}.$$

  Then the Magnus terms of the supersequence satisfy

  $$\boxed{h_m(T) = \frac{1}{T}\sum_{j=1}^{M} \tau_j h_m^{(j)}, \qquad m \in \{0, 1, \ldots, 2p\}.}$$

  In other words, all cross terms between subsequences vanish up to order $2p$.

  Proof: Since the first p terms vanish, each subsequence has

  $$\bar{H}_j(\tau_j, \lambda) = \sum_{m=p}^{\infty} \lambda^{m+1} h_m^{(j)} = \mathcal{O}(\lambda^{p+1}).$$

Now consider the BCH expansion of

$$\tilde{U}(T,\lambda) = \prod_{j=1}^{M} e^{-i\tau_j \bar{H}_j(\tau_j,\lambda)}.$$

Every term involving commutators between different subsequences contains at least two factors of $\bar{H}_j$, and is therefore at least $\mathcal{O}(\lambda^{2p+2})$.

Thus, up to this order, the logarithm of the product is simply the sum of the individual exponents:

$$-iT\bar{H}(T,\lambda) = -i\sum_{j=1}^{M} \tau_j \bar{H}_j(\tau_j,\lambda) + \mathcal{O}(\lambda^{2p+2}).$$

Substituting the Magnus expansions,

$$-iT\sum_{m=0}^{\infty} \lambda^{m+1} h_m(T) = -i\sum_{m=0}^{\infty} \lambda^{m+1} \sum_{j=1}^{M} \tau_j h_m^{(j)} + \mathcal{O}(\lambda^{2p+2}).$$

Equating powers of λ^{m+1} for $m \le 2p$ gives the final result $h_m(T) = \sum_{j=1}^{M} \tau_j h_m^{(j)}/T$.

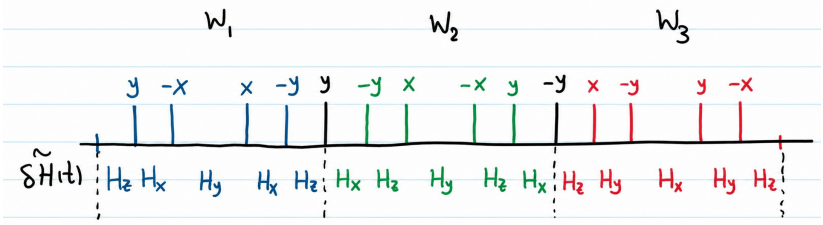
- *Special case:* If only the zeroth-order terms vanish within each subsequence, i.e. $h_0^{(j)} = 0$ for all j, then

$$\boxed{\begin{aligned} h_0(T) &= 0, \\ h_1(T) &= \frac{1}{T}\sum_{j=1}^{M} \tau_j h_1^{(j)}, \\ h_2(T) &= \frac{1}{T}\sum_{j=1}^{M} \tau_j h_2^{(j)}. \end{aligned}}$$

This provides a systematic way to construct higher-order decoupling sequences. For example, we can combine subsequences whose zeroth-order terms vanish, and choose them such that $\sum_{j=1}^{M} \tau_j h_1^{(j)} = 0$. The supersequence then satisfies $h_0(T) = h_1(T) = 0$.

- *Example:* Recall that the WAHUHA sequence suppresses dipolar interactions to first order, with $h_2 \neq 0$ in general. We can use the principles developed in this section to engineer a supersequence out of WAHUHA blocks that average out dipolar interactions to third order instead.

Consider the supersequence made out of three WAHUHA blocks (and two extra $\pi/2$ pulses)



where the durations are $\tau, \tau, 2\tau, \tau, \tau$ in each subsequence. The extra $\pi/2)_{\pm y}$ pulses rotate the middle WAHUHA block so that its toggling-frame Hamiltonian evolves as H_x, H_z, H_y, H_z, H_x, while ensuring the control unitary returns to $\mathbb{1}$ afterwards.

Each subsequence satisfies $h_0^{(j)} = h_1^{(j)} = 0$, just like the original WAHUHA sequence. By applying the same

calculation as in Eq. (8), we obtain

$$h_2^{(1)} = -\frac{\tau^2}{18}[H_z - H_y, [H_z, H_x]],$$
$$h_2^{(2)} = +\frac{\tau^2}{18}[H_x - H_y, [H_z, H_x]],$$
$$h_2^{(3)} = +\frac{\tau^2}{18}[H_z - H_x, [H_z, H_x]].$$

Since $h_0^{(j)} = h_1^{(j)} = 0$, pulse-cycle decoupling with $p = 2$ applies, and the second-order Magnus term of the entire supersequence is

$$\begin{aligned} h_2 &= \frac{1}{3}\left(h_2^{(1)} + h_2^{(2)} + h_2^{(3)}\right) \\ &= \frac{\tau^2}{54}[-(H_z - H_y) + (H_x - H_y) + (H_z - H_x), [H_z, H_x]] \\ &= 0. \end{aligned}$$

Furthermore, $h_3^{(j)} = 0$ for each subsequence by time symmetry, so pulse-cycle decoupling also gives $h_3 = 0$. Therefore,

$$\boxed{h_0 = h_1 = h_2 = h_3 = 0.}$$

So by combining three phase cycled WAHUHA blocks (and two extra rotations), we extended the dipolar averaging of the sequence to third order.

Appendix A Average Generators Beyond Hamiltonian Dynamics

- Consider a linear first-order matrix initial value problem $\dot{V}(t) = G(t)V(t)$, with initial value $V(0) = \mathbb{1}$, and solution $V(T) = \mathrm{Texp}\left[\int_0^T dt\, G(t)\right]$. When $G(t)$ is not Hermitian, can we still write $V(T)$ using an average generator?

- *Claim:* The inverse of $V(T)$ is the solution to $\dot{V}^{-1}(t) = -V^{-1}(t)G(t)$ with $V^{-1}(0) = \mathbb{1}$.
  Proof: Differentiate the product:

$$\begin{aligned} \frac{d}{dt}\big(V^{-1}(t)V(t)\big) &= \dot{V}^{-1}(t)V(t) + V^{-1}(t)\dot{V}(t) \\ &= -V^{-1}(t)G(t)V(t) + V^{-1}(t)G(t)V(t) = 0. \end{aligned}$$

  Since $V^{-1}(0)V(0) = \mathbb{1}$, we have $V^{-1}(t)V(t) = \mathbb{1}$, as desired.

- Every invertible complex matrix has a matrix logarithm [1], meaning that an average generator $\bar{G}(T)$ exists:

$$\boxed{\bar{G}(T) = \frac{1}{T}\ln V(T), \qquad V(T) = e^{T\bar{G}(T)}}.$$

- For example, Markovian evolution of an open quantum system is governed by the master equation

$$\dot{\rho}(t) = \mathscr{L}_t\left[\rho(t)\right],$$

  where $\mathscr{L}_t : \mathcal{L}(\mathcal{H}) \to \mathcal{L}(\mathcal{H})$ is the Liouvillian. The solution has the form

$$\rho(T) = \underbrace{\mathrm{Texp}\left[\int_0^T dt\, \mathscr{L}(t)\right]}_{V(T)}\rho(0),$$

  so the same argument gives $V(T) = e^{T\bar{\mathscr{L}}(T)}$, where $\bar{\mathscr{L}}(T)$ is the average Liouvillian.

Appendix B Dyson-Type Simplex Integrals

- *Claim:* Let $f : [0, a] \to \mathbb{C}$. The nested integral satisfies

$$\int_0^a dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-1}} dx_n \, f(x_1) \cdots f(x_n) = \frac{1}{n!} \left(\int_0^a dx \, f(x) \right)^n . \tag{9}$$

Proof: Define the antiderivative $F(x) = \int_0^x dt \, f(t)$. Integrating from the inside out, we have

$$\begin{aligned}
\int_0^a dx_1 \cdots \int_0^{x_{n-1}} dx_n \, f(x_1) \cdots f(x_n) &= \int_0^a dx_1 \, f(x_1) \cdots \int_0^{x_{n-2}} \underbrace{dx_{n-1} \, f(x_{n-1})}_{du} \underbrace{F(x_{n-1})}_{u} \\
&= \frac{1}{2} \int_0^a dx_1 \, f(x_1) \cdots \int_0^{x_{n-3}} \underbrace{dx_{n-2} \, f(x_{n-2})}_{du} \underbrace{F^2(x_{n-2})}_{u^2} \\
&= \frac{1}{2 \times 3} \int_0^a dx_1 \, f(x_1) \cdots \int_0^{x_{n-4}} \underbrace{dx_{n-3} \, f(x_{n-3})}_{du} \underbrace{F^3(x_{n-3})}_{u^3} \\
&\vdots \\
&= \frac{1}{(n-1)!} \int_0^a \underbrace{dx_1 \, f(x_1)}_{du} \underbrace{F^{n-1}(x_1)}_{u^{n-1}} \\
&= \frac{1}{n!} F^n(a) = \frac{1}{n!} \left(\int_0^a dx \, f(x) \right)^n .
\end{aligned}$$

$$\boxed{\int_0^a dx_1 \cdots \int_0^{x_{n-1}} dx_n \, f(x_1) \cdots f(x_n) = \frac{1}{n!} \left(\int_0^a dx \, f(x) \right)^n .}$$

- *Special case:* $\int_0^a dx_1 \cdots \int_0^{x_{n-1}} dx_n = a^n / n!$.

Appendix C Bare Magnus Expansion

- The same exact arguments developed here can be used outside the context of Hamiltonians, propagators, etc. For arbitrary finite-dimensional matrices $A_1, \ldots, A_n$:

$$\begin{aligned}
\prod_{k=1}^{n} e^{\lambda A_k} = \exp \Bigg[& \lambda \sum_{i=1}^{n} A_i + \frac{\lambda^2}{2} \sum_{i>j} [A_i, A_j] \\
& + \frac{\lambda^3}{12} \sum_{i>j} \big([A_i, [A_i, A_j]] + [A_j, [A_j, A_i]] \big) \\
& + \frac{\lambda^3}{6} \sum_{i>j>k} \big([A_i, [A_j, A_k]] + [A_k, [A_j, A_i]] \big) + \mathcal{O}(\lambda^4) \Bigg].
\end{aligned} \tag{10}$$

- *Special case:* BCH formula

$$e^{\lambda A_2} e^{\lambda A_1} = \exp \Big[\lambda (A_1 + A_2) + \frac{\lambda^2}{2} [A_2, A_1] + \frac{\lambda^3}{12} \big([A_2, [A_2, A_1]] + [A_1, [A_1, A_2]] \big) + \mathcal{O}(\lambda^4) \Big].$$

Appendix D Derivative of Matrix Exponential

- Let $X(t)$ be an operator-valued function of time. Its exponential has derivative

$$\frac{d}{dt} e^{X(t)} = \int_0^1 ds \, e^{(1-s)X(t)} \dot{X}(t) e^{sX(t)}. \tag{11}$$

- *Proof:* Define

$$F(t,s) \equiv e^{-sX(t)} \frac{\partial}{\partial t} e^{sX(t)}.$$

Now,

$$\frac{\partial}{\partial s} F(t,s) = -X(t) e^{-sX(t)} \frac{\partial}{\partial t} e^{sX(t)} + e^{-sX(t)} \frac{\partial}{\partial t} \left(X(t) e^{sX(t)} \right) = e^{-sX(t)} \dot{X}(t) e^{sX(t)}.$$

Integrating over s gives

$$\underbrace{F(t,1)}_{e^{-X(t)} \frac{d}{dt} e^{X(t)}} - \underbrace{F(t,0)}_{0} = \int_0^1 ds \, \frac{\partial}{\partial s} F(t,s),$$

$$e^{-X(t)} \frac{d}{dt} e^{X(t)} = \int_0^1 ds \, e^{-sX(t)} \dot{X}(t) e^{sX(t)}.$$

Multiplying by $e^{X(t)}$ gives

$$\boxed{\frac{d}{dt} e^{X(t)} = \int_0^1 ds \, e^{(1-s)X(t)} \dot{X}(t) e^{sX(t)}}.$$

- This is the same as the first-order term in a Dyson series! For the perturbed Hamiltonian $H(\lambda) = H_0 + \lambda \delta H$, the Taylor series gives

$$\begin{aligned} e^{-iTH(\lambda)} &= e^{-iTH(0)} + \lambda \left. \frac{d}{d\lambda} e^{-iTH(\lambda)} \right|_{\lambda=0} + \mathcal{O}(\lambda^2) \\ &= e^{-iTH_0} - i\lambda e^{-iTH_0} \int_0^T dt \, e^{itH_0} \delta H e^{-itH_0} + \mathcal{O}(\lambda^2). \end{aligned}$$

Alternative Derivation

- The trick is to first see that

$$\begin{aligned} e^{X(t+\Delta t)} - e^{X(t)} &= e^{(1-1)X(t)} e^{1X(t+\Delta t)} - e^{(1-0)X(t)} e^{0X(t+\Delta t)} \\ &= e^{(1-s)X(t)} e^{sX(t+\Delta t)} \Big|_{s=0}^{s=1} \\ &= \int_0^1 ds \, \frac{d}{ds} \left(e^{(1-s)X(t)} e^{sX(t+\Delta t)} \right). \end{aligned}$$

The exponents commute with themselves for different s, so

$$\begin{aligned} e^{X(t+\Delta t)} - e^{X(t)} &= \int_0^1 ds \left(-X(t) e^{(1-s)X(t)} e^{sX(t+\Delta t)} + e^{(1-s)X(t)} X(t+\Delta t) e^{sX(t+\Delta t)} \right) \\ &= \int_0^1 ds \, e^{(1-s)X(t)} \big(X(t+\Delta t) - X(t) \big) e^{sX(t+\Delta t)} \\ &= \int_0^1 ds \, e^{(1-s)X(t)} \big(\dot{X}(t)\Delta t + \mathcal{O}(\Delta t^2) \big) e^{s(X(t) + \dot{X}(t)\Delta t + \mathcal{O}(\Delta t^2))} \\ &= \Delta t \int_0^1 ds \, e^{(1-s)X(t)} \dot{X}(t) e^{sX(t)} + \mathcal{O}(\Delta t^2). \end{aligned}$$

where we used $e^{s(X(t) + \dot{X}(t)\Delta t + \mathcal{O}(\Delta t^2))} = e^{sX(t)} + \mathcal{O}(\Delta t)$. Dividing by Δt gives

$$\frac{e^{X(t+\Delta t)} - e^{X(t)}}{\Delta t} = \int_0^1 ds \, e^{(1-s)X(t)} \dot{X}(t) e^{sX(t)} + \mathcal{O}(\Delta t).$$

Hence, taking $\Delta t \to 0$, we get

$$\boxed{\frac{d}{dt} e^{X(t)} = \int_0^1 ds \, e^{(1-s)X(t)} \dot{X}(t) e^{sX(t)}}.$$

- You can remember the formula as follows: If $[X(t_1), X(t_2)] = 0$ for all t_1, t_2, we would have

$$\frac{d}{dt}e^{X(t)} = \dot{X}(t)e^{X(t)} = e^{X(t)}\dot{X}(t).$$

For the general case, we have a formula that smoothly interpolates between these two orders:

$$\frac{d}{dt}e^{X(t)} = \int_0^1 ds\, e^{(1-s)X(t)}\dot{X}(t)e^{sX(t)}.$$

The integrand $e^{(1-s)X(t)}\dot{X}(t)e^{sX(t)}$ is $e^{X(t)}\dot{X}(t)$ at $s = 0$ and $\dot{X}(t)e^{X(t)}$ at $s = 1$.

Appendix E First Few Dyson and Magnus Conversion Formulas

- For short, we write $\mathcal{D}_n = \mathcal{D}_n(\delta\tilde{H}, T)$ and $h_m = h_m(T)$ here. The first three expansion terms are related as follows
- *Dyson to Magnus:*

$$\boxed{\begin{aligned} h_0 &= \frac{\mathcal{D}_1}{T}, \\ h_1 &= -\frac{i}{T}\left(\mathcal{D}_2 - \frac{1}{2}\mathcal{D}_1^2\right), \\ h_2 &= -\frac{1}{T}\left[\mathcal{D}_3 - \frac{1}{2}(\mathcal{D}_1\mathcal{D}_2 + \mathcal{D}_2\mathcal{D}_1) + \frac{1}{3}\mathcal{D}_1^3\right]. \end{aligned}}$$

- *Magnus to Dyson:*

$$\boxed{\begin{aligned} \mathcal{D}_1 &= Th_0, \\ \mathcal{D}_2 &= iTh_1 + \frac{T^2}{2}h_0^2, \\ \mathcal{D}_3 &= -Th_2 + \frac{iT^2}{2}(h_0h_1 + h_1h_0) + \frac{T^3}{6}h_0^3. \end{aligned}}$$

The general formulas and proofs are in Section 3.1 and Lemma 12.

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